Coordinate Geometry

ANSWER KEY

Section 1: Slopes, Intercepts & Lines

1 What is the slope of the line perpendicular to the line with equation $3x - 4y = 12$?

a. $\frac{4}{3}$

b. $-\frac{4}{3}$ ✓

c. $\frac{3}{4}$

d. $-\frac{3}{4}$

Rewrite in slope-intercept form: $4y = 3x - 12 \Rightarrow y = \frac{3}{4}x - 3$. The slope is $m_1 = \frac{3}{4}$. A perpendicular line has slope $m_2 = -\frac{1}{m_1} = -\frac{4}{3}$.

2 A line passes through the points $(2, -3)$ and $(6, 5)$. What is the y-intercept of this line?

a. -7 ✓

b. -5

c. -3

d. 1

Slope $m = \frac{5-(-3)}{6-2} = \frac{8}{4} = 2$. Using point-slope: $y - 5 = 2(x - 6) \Rightarrow y = 2x - 12 + 5 \Rightarrow y = 2x - 7$. The y-intercept is -7.

3 Which of the following points lies on the line that passes through $(1, 4)$ and has a slope of -3?

a. $(3, -2)$ ✓

b. $(2, 2)$

c. $(4, -4)$

d. $(0, 6)$

Line equation: $y - 4 = -3(x - 1) \Rightarrow y = -3x + 7$. Testing $(3, -2)$: $-3(3) + 7 = -9 + 7 = -2$ (true).

Section 2: Midpoints & Distance Formula

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4 What is the distance between the points $A(-3, 2)$ and $B(5, -4)$ in the standard (x, y) coordinate plane?

a. 8

b. 10 ✓

c. 12

d. 14

Use the distance formula: $d = \sqrt{(5-(-3))^2 + (-4-2)^2} = \sqrt{8^2 + (-6)^2} = \sqrt{64+36} = \sqrt{100} = 10$.

5 The midpoint of line segment $\overline{CD}$ is $M(4, -1)$. If point C has coordinates $(1, 5)$, what are the coordinates of point D?

a. $(7, -7)$ ✓

b. $(6, -6)$

c. $(5, -3)$

d. $(7, -6)$

Midpoint formula: $\frac{1+x_D}{2} = 4 \Rightarrow x_D = 7$, and $\frac{5+y_D}{2} = -1 \Rightarrow y_D = -7$. Thus $D(7, -7)$.

6 What is the equation of the perpendicular bisector of the segment connecting $(0, 2)$ and $(4, 6)$?

a. $y = -x + 6$ ✓

b. $y = -x + 4$

c. $y = x + 2$

d. $y = -2x + 8$

Midpoint is $(\frac{0+4}{2}, \frac{2+6}{2}) = (2, 4)$. Slope of segment is $\frac{6-2}{4-0} = 1$. Perpendicular slope is -1. Equation: $y - 4 = -1(x-2) \Rightarrow y = -x + 6$.

Section 3: Circles in the Coordinate Plane

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7 What are the center and radius of the circle given by $(x-5)^2+(y+2)^2=49$?

a. **Center** $(5,\ -2)$**, radius** 7 ✓

b. Center $(-5, 2)$, radius 7

c. Center $(5,\ -2)$, radius 49

d. Center $(-5, 2)$, radius 49

Standard circle form: $(x-h)^2+(y-k)^2=r^2$. Here $h=5$, $k=-2$, and $r=\sqrt{49}=7$.

8 A circle has the equation $x^2+y^2-6x+8y=0$. What is the radius of this circle?

a. 4

b. 5 ✓

c. 6

d. 10

Complete the square: $(x^2-6x+9)+(y^2+8y+16)=9+16 \Rightarrow (x-3)^2+(y+4)^2=25$. Radius $r=\sqrt{25}=5$.

9 Which of the following lines is tangent to the circle $x^2+y^2=25$ at the point $(3, 4)$?

a. $3x+4y=25$ ✓

b. $4x+3y=25$

c. $3x-4y=25$

d. $4x-3y=0$

Radius to $(3, 4)$ has slope $\frac{4}{3}$. Tangent line has perpendicular slope $-\frac{3}{4}$:
$y-4=-\frac{3}{4}(x-3) \Rightarrow 4y-16=-3x+9 \Rightarrow 3x+4y=25$.

Section 4: Parabolas & Conic Forms

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10 What is the vertex of the parabola defined by the equation $y = 2x^2 - 8x + 11$?

a. $(2, 3)$ ✓

b. $(2, 5)$

c. $(-2, 35)$

d. $(4, 11)$

x-coordinate of vertex: $x = -\frac{b}{2a} = -\frac{-8}{2(2)} = 2$. y-coordinate: $y = 2(2)^2 - 8(2) + 11 = 8 - 16 + 11 = 3$. Vertex is $(2, 3)$.

11 What is the equation of the axis of symmetry for the quadratic function $y = -3(x + 4)^2 - 7$?

a. $x = -4$ ✓

b. $x = 4$

c. $y = -7$

d. $x = -7$

In vertex form $y = a(x - h)^2 + k$, the axis of symmetry is the vertical line $x = h$. Here $h = -4$, so $x = -4$.

12 At how many points do the graphs of $y = x^2 - 4x + 5$ and $y = 2x - 4$ intersect?

a. 0

b. 1 ✓

c. 2

d. Infinitely many

Set equal: $x^2 - 4x + 5 = 2x - 4 \Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x - 3)^2 = 0$. Since there is exactly one solution ($x = 3$), they intersect at exactly 1 point.

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